

Ifeoma Precious Ogwu

ifeomalockwood@gmail.com | linkedin.com/in/ifeoma-ogwu/ | US Citizen

Education

Bachelor Of Science (B.S) in Electrical Engineering

The University of Tulsa, Overall GPA: 3.66, Major GPA: 3.85

Dec 2025

- Graduated Cum Laude
- Received Dean's Honor Roll (Fall 2022, Fall 2025) and President's Honor Roll (Spring 2023, Fall 2024, Spring 2025)
- Relevant Coursework: Electronics, Controls Theory, Power Systems, Electromagnetics, Circuits, Embedded Systems, Digital Systems and Design, Microprocessors, Signals and Systems, Computer Organization, Optical Networking

Experience

Distribution Engineer Intern, American Electric Power - Tulsa, Oklahoma

June 2025 - Dec 2025

- Worked in the distribution reliability group and supported CVR and DACR efforts
- Configured base cases for **400+** feeders in **CYMDIST** and exported datasets to SQL supporting Perfect Power initiatives
- Conducted **load allocation**, **ampacity studies**, **new load studies** and **short circuit studies** to inform protection planning
- Assisted commissioning of IntelliRupter reclosers and the configuring of protection settings and resetting modules, reducing outage resolution time in the field
- Performed coordination analyses in **CYMETCC** to optimize feeder performance and inform reliability improvements
- Created dashboards using **PI Vision** for displaying feeders, their devices and various status points for YFA and CVR efforts
- Conducted **Arc Flash** studies in CYMDIST per customer requests to gain analysis results on specified nodes

Business Analyst Intern, SageNet - Tulsa, Oklahoma

May 2024 - Aug 2024

- Published **100s** of SharePoint subsites used as runbooks for the purpose of a document repository project
- Decreased the gap between operational cost and net profit by 4% through new documentation system
- Created **Python** scripts to automate data migration from old database to Microsoft SharePoint
- Implemented Power Automate to automate workflow actions for the runbooks made in SharePoint
- Analyzed cybersecurity reports for data validation and segmentation purposes
- Utilized **Power BI** to create report dashboards for operational organization and data analysis
- Optimized Raspberry Pi boot sequence to load into browser in under 15 seconds

Robotics Researcher, The University of Tulsa

May 2024 - Aug 2024

- Designed **PCB** schematics and models for a controller to integrate with the Boston Dynamics' mobile robot: Spot
- Utilized KiCAD to create detailed schematics and multi-layer PCB layouts
- Researched and tested best components to use for the controller and best ways for production using **Raspberry Pi**
- Collaborated with a multidisciplinary team to ensure seamless integration with Spot's existing systems

Manufacturing Engineering Intern, Atento Capital | Access Optics - Tulsa, Oklahoma

May 2023 - July 2023

- Worked under Atento Capital for one of their portfolio companies, Access Optics, as a manufacturing engineering intern
- Built documentation libraries to support project-specific and continuous use to improve operational efficiency
- Created project reports, assembly processes and work instructions documentation
- Performed research for pricing and purchasing benefit
- Interpreted **technical drawings** and manufacturer specifications to support experienced engineers
- Refined engineering designs to incorporate new information and specifications

Projects

Ambient Light Energy Harvesting For Microcontroller | *ArduinoIDE, AEM10941, Solar cells, Arduino, Sensors*

- Engineered a self-sustaining IoT data logger using an **Arduino Pro Mini** and **AEM10941** harvester to evaluate solar extension of battery life
- Programmed firmware in **C++** to interface with BME280, DS3231, and MAX17048 sensors via **I2C and SPI**, implementing **deep sleep** algorithms
- Analyzed system power consumption and diagnosing points of issues
- Designed a custom energy harvesting PCB in **KiCad** utilizing the **BQ25570** boost charger to enable continuous harvesting

from input voltages as low as **100 mV** and correct load management.

Portable Gaming System | *STM32CubeIDE, STM32 Microcontroller, Sensors, GitHub, SPI, I2C, I2S, UART*

- Designed and implemented a portable gaming system featuring user-selectable games (tetris and rhythm game) via push buttons and interactive gameplay using an **STM32** microcontroller and TFT LCD breakout display
- Created a rhythm game integrating **MPU6050** sensor data for motion control, leveraging **Kalman filtering** to enhance tilt detection and precision
- Implemented a central GitHub codebase to handle version control and update code

ConocoPhillips Hackathon Participant | *Python, HTML, OpenLLM*

- Ranked 1st place in the competition by developing a cybersecurity-focused AI knowledge chatbot
- Developed a Python API for accessing **openLLM**, an open source large language model
- Designed and implemented the chatbot by integrating the API with **Flask** and **HTML** for frontend and backend development
- Demonstrated competence in Python and AI integration by successfully deploying the chatbot

Skills

Programming languages: MATLAB, Simulink, C, Python, MIPS and RISC-V assembly language, QB64, Verilog

Software: Microsoft Office, SolidWorks, KiCad, SharePoint, Power Automate, Power BI, Linux, CymDist, CymeTCC, IntelliLink

Hardware: Arduino, PCB design, Soldering, Design and Development, Embedded Systems, STM32 Microcontroller

Equipment: Oscilloscope, Multimeter, Circuit Elements, Function Generator, Breadboard, Raspberry Pi, 3D Printer